WM-DSLM: A Constrained Generation Architecture for Wealth Management AI

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Abstract

The wealth management industry has deployed large language models aggressively — every major wirehouse now operates an LLM copilot, and recent data show that 40% of registered investment adviser firms have implemented AI tools internally, yet 44% of those firms conduct no formal testing or validation of their outputs. This paper argues that the dominant paradigm — generate-then-validate — is structurally incompatible with fiduciary standards, introduces irreducible compliance risk, and produces legally indefensible provenance chains. We introduce WM-DSLM v1.0, a six-layer constrained generation architecture that shifts the compliance burden upstream into the generation process itself. The architecture advances six novel contributions: (1) Suitability-Constrained Decoding — encoding client suitability profiles as formal generation constraints applied at the token level, preventing unsuitable advice from being produced rather than catching it after; (2) Decision Contracts as machine-readable pre-inference governance; (3) a Policy-Aware Semantic Layer as always-injected versioned governance independent of model training; (4) Intent-Tiered Context Assembly mapping wealth management intents to distinct data requirements; (5) a Cryptographic Evidence Layer providing hash-chained, verifiable reasoning traces meeting SEC Rule 204-2 and FINRA record-keeping standards; and (6) a Bounded Autonomy Envelope structurally defining what the model may reason about per intent tier. The architecture is grounded in a systematic review of foundational financial LLMs, commercial wealth management AI deployments, and the compliance and governance literature through early 2026.

Keywords: domain-specific language models, wealth management, constrained generation, suitability enforcement, AI governance, decision contracts, cryptographic provenance, Reg BI, MiFID II, FCA Consumer Duty

JEL Classification: G20, G17, O33, K22

1. Introduction

Every major wealth management institution now operates a generative AI copilot. Morgan Stanley's AI Assistant has achieved 98% adoption among its advisor teams, drawing on over 100,000 proprietary research documents. JPMorgan's LLM Suite, initially rolled out to 50,000 employees in mid-2024, scaled to over 200,000 users within eight months — recognized by American Banker as its 2025 Innovation of the Year. Goldman Sachs launched its GS AI Assistant firmwide to approximately 46,000 employees in June 2025. The wealthtech ecosystem has absorbed over $200M in venture capital across 2025 and early 2026: Jump ($105M total), Zocks ($65M), Nevis ($40M from Sequoia Capital, ICONIQ, and Ribbit Capital). Envestnet delivers 25 million next-best-action suggestions daily.

Yet a 2026 report from the Cambridge Centre for Alternative Finance found that while 40% of investment adviser firms have implemented AI tools internally, 44% of those firms conduct no formal testing or validation of their AI outputs. This is not a peripheral compliance gap — it is the central architectural problem of wealth management AI. Every production system operates on the same paradigm: generate an output, then validate it for compliance. Compliance tools — Recordsure, Aveni, SAIFR — analyze advice after it has been generated. Guardrail frameworks intercept outputs before delivery. Origin's 138-check Compliance Gateway validates at the point of output. We refer to this pattern throughout as the *generate-then-validate paradigm*.

Generate-then-validate has three fundamental deficiencies in a fiduciary context. First, it is architecturally equivalent to pre-DevOps software testing — compliance applied after the artifact is produced, not during production. Second, it produces legally indefensible provenance: Limeglass (2025) documents that chain-of-thought explanations 'cannot be treated as audit evidence' because they are 'post-hoc reconstructions, not traces.' Third, LLM non-determinism means the same query can produce materially different advice at different times — structurally incompatible with Reg BI's documentation requirements and MiFID II's consistency obligations.

WM-DSLM v1.0 introduces a constrained generation architecture that reverses this paradigm. Rather than generating advice and validating it, WM-DSLM encodes suitability, policy, and regulatory constraints into the generation process itself. The compliance burden moves upstream: unsuitable advice cannot be produced because the generation architecture prevents it.

2. Related Work and Positioning

2.1 Foundational Financial LLMs

The domain-specific financial LLM literature centers on corpus-trained models optimized for NLP tasks. BloombergGPT (Wu et al., 2023) established the benchmark with a 50B parameter model trained on 363B tokens of Bloomberg financial data. FinGPT (Yang et al., 2023) democratized the approach via LoRA fine-tuning at approximately $300 per run. InvestLM (Yang, Tang, and Tam, 2023) fine-tuned on CFA exam questions and SEC filings. PIXIU/FinMA (Xie et al., NeurIPS 2023) provided the FLARE evaluation benchmark. None address inference-time suitability enforcement, regulatory constraint integration, or provenance architecture.

2.2 Applied Research on LLM Financial Advice

Lo and Ross (2024, Harvard Data Science Review) tested GPT-4 across financial domain knowledge and RAG-augmented ethical guidance, finding reasonable performance with persistent behavioral biases. Fieberg et al. (2024, CESifo WP 11666) elicited portfolio recommendations from 32 LLMs across 64 investor profiles, finding suitable advice with no gender discrimination but persistent home bias. Takayanagi et al. (2025a) found that users preferred extroverted AI personas despite receiving objectively worse advice. None of this work addresses the architecture of constraint enforcement.

2.3 Compliance and Governance Systems

Rainbird (2025) published on deterministic knowledge-graph inference for FCA suitability guardrailing — the closest existing work — but operates as post-generation validation. The Agent Behavioral Contracts framework (Bhardwaj, 2026) established Design-by-Contract principles for AI agents as C = (P, I, G, R): Preconditions, Invariants, Governance policies, Recovery mechanisms. WM-DSLM instantiates this framework for wealth management with direct regulatory binding. CNFinBench (arXiv:2512.09506, 2024) defined behavioral specifications for financial LLM safety but provides no enforcement mechanism. The FINOS AI Governance Framework (2025) identifies 'retrieval-response disconnect' as a key failure mode but remains principles-based without architectural prescription.

2.4 Positioning

Prior Work / System	Focus	WM-DSLM Relationship
BloombergGPT, FinGPT, InvestLM	Fine-tuning on financial corpora for NLP tasks	WM-DSLM adds the constraint enforcement layer all such models lack
Lo & Ross (2024); Fieberg et al. (2024)	Evaluating what LLMs produce for advice	WM-DSLM defines how to structurally prevent unsuitable outputs
Rainbird (2025)	Knowledge-graph guardrails for FCA suitability	Rainbird validates post-generation; WM-DSLM constrains during generation
ABC Framework (Bhardwaj, 2026)	Design-by-Contract for AI agents — general purpose	WM-DSLM is a wealth management instantiation with direct regulatory binding
Guardrails AI, NeMo, Bedrock	Generic output validation frameworks	Domain-agnostic; no suitability semantics or financial regulatory taxonomy
Morgan Stanley, Jump, Zocks	Advisor workflow productivity	Operational only; no published architecture for constrained generation
FINOS AI Governance Framework (2025)	General financial AI governance principles	Principles-based; WM-DSLM provides the implementation specification

Table 1. WM-DSLM v1.0 positioning against existing work.

3. The Generate-Then-Validate Paradigm and Its Limits

The generate-then-validate paradigm is commercially dominant and technically mature. It is also architecturally insufficient in regulated advice contexts for three reasons.

> *Reason 1: Validation is not prevention.* A compliance filter that catches 95% of unsuitable outputs still produces unsuitable advice in 5% of cases. In financial services, where a single unsuitable recommendation can trigger regulatory action, best-effort filtering is not a fiduciary standard. The Cambridge CCAF (2026) finding that 44% of AI-using RIA firms conduct no output validation at

all underscores the severity of this gap. WM-DSLM proposes the equivalent of shift-left testing — embedding compliance at generation, not after.

Reason 2: Post-hoc explanations are not audit evidence. Chain-of-thought prompting produces reasoning narratives generated after the output. Limeglass (2025) documents these are 'post-hoc reconstructions, not traces.' SEC Rule 204-2(a)(7) requires RIAs to maintain records of the basis for recommendations. The SEC's FY2026 Examination Priorities (released November 17, 2025) explicitly scrutinize AI governance and flag 'AI washing' — misleading claims about AI capabilities — as an examination priority. A CoT explanation with no verifiable binding to the data inputs used fails this standard.

Reason 3: LLM non-determinism is incompatible with regulatory consistency. The same prompt and client profile can produce materially different advice at different times due to temperature sampling. MiFID II Article 25 requires suitability assessments to be consistent and reproducible. Suitability-Constrained Decoding addresses this by encoding the client's profile as structural constraints on generation, producing deterministic constraint-satisfaction for critical output dimensions.

4. Regulatory Context: Principles Without Architecture

Three regulatory regimes govern AI-generated financial advice. All three are undergoing significant change as of early 2026, yet all remain principles-based with no binding architectural requirements for AI advice systems.

Regime	Status as of Early 2026	Architectural Gap
Reg BI (US)	SEC withdrew the Predictive Data Analytics rule on June 12, 2025. FY2026 Exam Priorities (Nov 17, 2025) explicitly scrutinize AI governance, 'AI washing,' and automated advice platforms. 40% of RIA firms have AI tools but 44% conduct no output validation (CCAF, 2026). US Treasury wrapped up a cross-agency AI risk management initiative in February 2026.	No binding technical standards for AI advice generation. No specification of what constitutes an adequate audit trail. Examination scrutiny is increasing, but architectural requirements do not exist.
MiFID II (EU)	ESMA confirmed (May 2024) suitability requirements apply to AI. Investment advice is NOT classified as high-risk under the EU AI Act — only credit scoring and insurance are covered. Oxford Law Blog (February 2026) calls for 'MiFID III' specifically addressing AI obligations.	No binding rules on AI model design, data governance, explainability, or traceability in advice contexts. The EU AI Act's omission of investment advice from the high-risk category creates a material regulatory vacuum.

Regime	Status as of Early 2026	Architectural Gap
FCA Consumer Duty (UK)	FCA launched the Mills Review on January 27, 2026, led by Sheldon Mills, examining AI's long-term impact on retail financial services through 2030. Reporting to the FCA Board in summer 2026. First cohort of firms joined AI Live Testing in October 2025; second cohort launching April 2026. UK Treasury Select Committee urged the FCA to publish binding AI guidance by end of 2026.	Mills Review is explicitly outcomes-based and technology-neutral — no architectural requirements anticipated. Treasury Select Committee criticism suggests the FCA's 'wait and see' approach may be exposing consumers to risk.

Table 2. Regulatory framework status as of early 2026.

The pattern across all three regimes is consistent: regulatory scrutiny is intensifying rapidly, but technical requirements remain absent. WM-DSLM is designed to define the architecture that binding regulation will eventually require. Firms that implement constrained generation architectures now will have a compliance and examination advantage when prescriptive requirements arrive.

5. WM-DSLM v1.0 Architecture: Six Layers

WM-DSLM v1.0 consists of six layers operating in sequence at inference time. Layers 1 through 4 establish the governance and context framework. Layer 5 — Suitability-Constrained Decoding — is the primary architectural contribution. Layer 6 provides cryptographic provenance.

#	Layer	Primary Function
1	Intent Classification	Classify query into intent tier; invoke the appropriate Decision Contract
2	Decision Contract Validation	Validate inputs for presence, freshness, and permissions; block if conditions not met
3	Policy-Aware Semantic Layer	Inject versioned canonical definitions and firm policy constraints at runtime — not retrieved
4	Bounded Autonomy Envelope	Structurally specify what the model may and may not reason about per intent tier
5	Suitability-Constrained Decoding	Encode client suitability profile as formal generation constraints applied during token generation — the core contribution
6	Cryptographic Evidence Layer	Produce hash-chained, verifiable provenance record concurrent with inference

Table 3. WM-DSLM v1.0 six-layer inference architecture.

5.1 Layers 1–4: Governance and Context Framework

Layer 1: Intent Classification. Queries are classified into five intent tiers (T1 ADVICE_PERSONALIZED through T5 COMPLIANCE_CHECK) before any data is retrieved. The tier determines which Decision Contract governs the interaction and which suitability constraints apply.

Layer 2: Decision Contracts. A Decision Contract (DC) is a machine-readable, versioned specification defining pre-conditions, required inputs with freshness and permission rules, hard output constraints, and conditions that halt inference entirely. The reference implementation DC-NBC-ADVISOR-0.1 governs next-best conversation with 10 required inputs, 3 blocking conditions, and 5 flag conditions. This instantiates the ABC framework (Bhardwaj, 2026) with binding to Reg BI, MiFID II Article 25, and FCA Consumer Duty.

Layer 3: Policy-Aware Semantic Layer. A versioned, firm-governed set of canonical definitions injected at inference time — not retrieved. When 'suitable' appears in a query, the model applies the regulatory definition consistent with FINRA Rule 2111, Reg BI, and MiFID II Article 25, not the colloquial meaning. Policy changes require a version increment, not model retraining.

Layer 4: Bounded Autonomy Envelope. A structural specification of what the model may and may not reason about per intent tier, enforced as contract schema rather than prompt instruction. Components: permitted output categories (generic asset classes only), prohibited outputs (specific product names, tax advice, legal advice), required citation fields, mandatory flag requirements, and data completeness thresholds for confidence ratings.

5.2 Layer 5: Suitability-Constrained Decoding

Suitability-Constrained Decoding (SCD) encodes a client's suitability profile as formal generation constraints applied during the token generation process — not as post-hoc validation. This is the core contribution of WM-DSLM v1.0.

Theoretical Foundation.

Constrained decoding is an established NLP technique: grammar-constrained generation (Guidance, LMQL, Outlines), logit masking for structured outputs, and type-constrained generation all demonstrate that token-level constraints are technically tractable. WM-DSLM applies these techniques to financial suitability specifications — a novel application not previously published in the financial AI literature.

Formally, standard generation samples each token from P(y_t | context). SCD modifies this to:

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P_SCD(y_t | context) = P(y_t | context) x [y_t satisfies C_client] / Z
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where C_client is the client's Suitability Constraint Specification and Z is a normalizing constant. Tokens whose generation would violate a suitability constraint receive probability zero.

Suitability Constraint Specification.

Dimension	Source	Example Constraint
Risk tolerance bounds	Documented risk assessment	If risk = Conservative: P(tokens recommending equities > 30% of portfolio) = 0
Eligibility gates	Account type, investor status	If accredited_investor = False: P(tokens referencing alternative investments) = 0

Dimension	Source	Example Constraint
Concentration limits	IPS or regulatory defaults	If single_security > 20%: P(tokens recommending increasing that position) = 0
Jurisdiction constraints	Client address, regulatory regime	If jurisdiction = EU: P(tokens referencing products outside PRIIPS scope) = 0
Time horizon guards	Client-stated horizon	If time_horizon < 3 years: P(tokens recommending illiquid alternatives) = 0

Table 4. Suitability Constraint Specification — five constraint dimensions.

Implementation Modes.

Hard logit masking (strongest): Constrained tokens receive probability = 0. Used for binary eligibility constraints. Requires white-box model access.

Soft penalty (probabilistic): Constrained tokens receive reduced probability. Used for risk tolerance bounds and concentration limits. Requires white-box access.

Structured output schema (widest compatibility): Output constrained to a JSON schema with field-level validators. Compatible with commercial API-access models. Immediately deployable without model access.

5.3 Layer 6: Cryptographic Evidence Layer

The Evidence Layer produces a hash-chained provenance record concurrent with — not after — inference. This directly addresses the Limeglass finding that current AI audit trails are legally indefensible, and the SEC FY2026 Exam Priorities' scrutiny of AI governance documentation.

Field	Content	Regulatory Purpose
input_hash	SHA-256 of assembled context block	Verifies exactly what data the model saw
constraint_hash	SHA-256 of SCS applied	Verifies which suitability constraints were active
semantic_layer_version	Semver string	Establishes which definitions governed the output
contract_id + version	e.g., DC-NBC-ADVISOR-0.1	Documents which governance contract applied — per Reg BI best interest standard
retrieval_record	List of {source_id, doc_version, retrieved_at}	Per Reg BI 204-2(a)(7): basis for recommendation with timestamps
output_hash	SHA-256 of generated output	Verifies output has not been modified post-generation
chain_hash	SHA-256(input + constraint + output hashes)	Cryptographic linkage — post-hoc modification is detectable

Field	Content	Regulatory Purpose
suitability_pass	Boolean + constraint evaluation log	Documents suitability evaluation for Reg BI and MiFID II Article 25

Table 5. Cryptographic Evidence Layer — record schema.

6. Novel Claims

WM-DSLM v1.0 advances six claims of architectural novelty, each validated against the existing literature through early 2026.

Claim 1: Suitability-Constrained Decoding. The first formalization of inference-time suitability enforcement using constrained decoding applied to financial suitability specifications. All prior compliance work operates post-generation. SCD prevents rather than catches unsuitable advice.

Claim 2: Decision Contracts as Pre-Inference Governance. Machine-readable, versioned contracts that validate data conditions and can halt model reasoning before output is generated — extending data contract theory (Jones, 2023) from pipeline governance to the reasoning step itself, with binding to Reg BI, MiFID II Article 25, and FCA Consumer Duty.

Claim 3: Policy-Aware Semantic Layer as Versioned Governance Artifact. Canonical regulatory definitions separated from model training and prompt instruction into independently versioned artifacts injected at runtime. Policy changes require a version increment, not model retraining.

Claim 4: Intent-Tiered Context Assembly. A formal taxonomy of wealth management advice intents mapping to distinct data requirements, freshness rules, permission checks, and constraint sets. No prior financial AI publication formalizes intent-to-context assembly specifications.

Claim 5: Cryptographic Evidence Layer for Regulatory Provenance. A hash-chained provenance record produced concurrent with inference — not as post-hoc explanation — designed to satisfy SEC Rule 204-2(a)(7), FINRA Rules 17a-3 and 17a-4, and equivalent EU and UK record-keeping requirements. Directly answers the SEC FY2026 examination scrutiny of AI governance documentation.

Claim 6: The Constrained Generation Paradigm for Regulated AI Advice. A formal articulation of the paradigm shift from generate-then-validate to constrained generation, providing the theoretical basis for evaluating all wealth management AI architectures against a principled compliance standard.

7. Worked Example: Concentrated Position Scenario

The following traces a T2 (ANALYSIS_PORTFOLIO) query through all six layers, illustrating how Suitability-Constrained Decoding prevents an unsuitable recommendation rather than catching it after generation.

Parameter	Value
Client	Raj Patel, 48, Moderate-Aggressive risk tolerance
Holdings	72% equity; 22% concentrated in single tech stock (RSU vesting)
IPS constraint	Single-security concentration limit: 20% of portfolio
Query	'What should I do with my RSU vest next month?'
SCS loaded	P(tokens recommending hold or increase of single_security) = 0, because 22% > IPS limit of 20%

Layer	Action	Result
L1: Intent	T2 ANALYSIS_PORTFOLIO	DC-CONC-DIV-0.1 invoked
L2: Decision Contract	Holdings validated; concentration flag triggered (22% > 20%)	CONCENTRATION_RISK flag raised; inference proceeds with flag active
L3: Semantic Layer	Injected: 'concentrated_position' defined as single-security > 20% per regulatory definition	Canonical constraint grounded in regulatory definition, not colloquial usage
L4: Bounded Autonomy	BAE requires addressing IPS breach; prohibits specific product names	Constraint envelope applied before generation begins
L5: SCD	Hard logit mask: tokens recommending hold or increase of flagged position receive P = 0	Model cannot output a hold or increase recommendation. Diversification is the only available output space.
L6: Evidence Layer	input_hash, constraint_hash, output_hash, chain_hash generated concurrent with generation	Verifiable provenance record. Post-hoc modification is detectable.

Table 6. Layer-by-layer execution trace — RSU concentration scenario.

8. Evaluation Framework

Five metrics are proposed for evaluating constrained generation architectures in wealth management, designed to map directly to SEC FY2026 examination criteria and MiFID II Article 25 suitability assessment requirements.

Metric	Definition	Target
Constraint Satisfaction Rate	Percentage of outputs fully satisfying all active SCS constraints	>= 99.5% for hard constraints
Constraint Completeness	Percentage of applicable suitability rules encoded in SCS versus IPS and regulatory source	>= 95% for Reg BI Tier 1 constraints
Evidence Chain Integrity	Percentage of outputs with valid, unbroken hash chains linking input, constraint, and output	100% — any gap is a compliance failure
Provenance Recall	Percentage of retrieved documents and data sources captured in the evidence record	>= 99%

Metric	Definition	Target
Regulatory Coverage	Percentage of applicable regulatory requirements with corresponding SCS constraints for T1 and T5 intents	>= 90%

Table 7. WM-DSLM v1.0 proposed evaluation metrics.

9. Limitations and Future Work

- *White-box access requirement.* Hard logit masking requires access to model logits, limiting full SCD to self-hosted or white-box models. Structured output schema mode is available for commercial APIs but provides weaker guarantees.
- *SCS completeness challenge.* Encoding all applicable suitability rules requires expert translation of regulatory text into machine-readable constraints. The Compliance-to-Code framework (arXiv:2505.19804) and neuro-symbolic approaches (arXiv:2601.06181) offer promising automation directions.
- *Dynamic regulatory adaptation.* When regulations change, SCS must be updated manually. The FCA Mills Review (January 2026) and the anticipated FCA binding guidance by end of 2026 will require version updates to the Semantic Layer and SCS. Automated regulatory text-to-SCS translation remains an open research problem.
- *Fiduciary duty encoding.* The duty of loyalty remains a human judgment dimension that cannot be fully encoded as generation constraints. SCD addresses suitability but not the full fiduciary standard. The SEC FY2026 Exam Priorities' focus on 'AI washing' underscores that claims about AI fiduciary compliance require careful scoping.
- *Empirical validation.* This is a formal architecture specification. Empirical validation against real advisory scenarios, regulatory exam simulations, and financial safety benchmarks (CNFinBench, FLARE) is required before performance claims can be made.

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